

Bangladesh University of Business and Technology (BUBT)

Faculty of Engineering and Applied Sciences

Department of Mathematics and Statistics

Program: B. Sc. Engineering in CSE

Intake/Section: 53

Course Title: Complex Variable and Statistics

Course Code: MAT 231

Fall 2025 Assignment-2

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Data Arrangement and Representation

1. The profits (in lakhs of Tk's) of 30 Bangladeshi companies for the year 2005-2006 are given below:

18 16 23 37 35 49 63 65 55 45 58 57 69 20
22 35 42 37 42 48 53 49 65 39 48 67 25 29
58 65

Classify the above data taking a suitable class interval.

2. The following are the marks of the 30 students in statistics. Prepare a frequency distribution taking a suitable class interval.

12 33 23 25 18 35 37 49 54 51 37 15 27 33
42 45 47 55 69 65 63 46 29 18 37 45 46 59
29 55.

3. Form a frequency distribution for the following data-

67 34 36 48 49 31 61 34 43 45 38 32 27 61
29 47 36 50 46 30 46 32 30 33 45 49 48
41 53 36 37 37 47 62 46 50 28 35 35 38
36 47 43 34 62 69 50 28 44 43 60 39.

4. The funds flow of Goodwill India Ltd from 1991-92 to 1995-96 are given below:

Year	Funds Flow (Rs.Crores)
1991-1992	70
1992-1993	90
1993-1994	160
1994-1995	100
1995-1996	60

Represent this data by a suitable bar diagram.

5. Draw the Bar Diagram and pie chart from the following data-

Division	Rain fall (mm)
Dhaka	1540
Chittagong	2260
Khulna	1159
Rajshahi	1192
Sylhet	3568
Barisal	1200

6. Brian asks 60 people what their favorite color is and separates the answers into 5 categories. His results are shown in the table below. Draw a **pie chart** to display Brian's results.

Colour	Red	Blue	Green	Yellow	Other
Frequency	10	13	24	5	8

7. Mr. X asked his 20 friends' which types of movies they liked very much. The following data was found from his survey.

<i>Table: Favorite Type of Movie</i>				
Comedy	Action	Romance	Drama	Sci Fi
4	5	6	1	4

Show the above data with a **pie chart**.

8. Represent the following data by a histogram:

Size Class	Frequency	Size Class	Frequency
1-10	5	51-60	10
11-20	11	61-70	8
21-30	19	71-80	6
31-40	21	81-90	3
41-50	16	91-100	1

9. Draw the histogram from the following data-

Class Interval	Frequency
14.5 - 19.5	4
19.5 - 24.5	4
24.5 - 29.5	4
29.5 - 34.5	8
34.5 - 39.5	4
39.5 - 44.5	10
44.5 - 49.5	2
49.5 - 54.5	3

The measure of Central Tendency

10. The following are the figures of profits earned by 1500 companies during 1999-2000.

Profits (Tk.lakhs)	No. of companies	Profits (Tk.lakhs)	No. of companies
200-400	200	1000-1200	150
400-600	300	1200-1400	100
600-800	450	1400-1600	80
800-1000	200	1600-1800	20

Calculate the **average profits** for all companies.

11. Find the mean of the following frequency distribution:

Class	50-60	60-70	70-80	80-90	90-100	100-110	110-120
Frequency	15	25	50	100	60	30	20

12. The following table gives the monthly income of 12 families in a village.

H.No	Monthly income (Thousand TK)	H.No	Monthly income (Thousand TK)
1	587	7	805
2	693	8	907
3	595	9	763
4	780	10	865
5	840	11	768
6	760	12	894

Calculate the median income.

13. 1500 workers are working in an industrial establishment. Their age is classified as follows:

Age (yrs)	No. of workers	Age (yrs)	No. of workers
18-22	120	38-42	184
22-26	125	42-46	162
26-30	280	46-50	86
30-34	260	50-54	75
34-38	155	54-58	53

Calculate the median age.

14. The heights (in cm) of 11 players of a team are as follows: 160, 158, 158, 159, 160, 160, 162, 165, 166, 167, 170. Find the median of the above data.
15. If the median of 17, 13, 10, 15, x happens to be the integer x then find x .
16. Find the median of the first five odd integers. If the sixth odd integer is also included, find the difference of medians in the two cases.
17. The following are scores obtained by 11 players in a cricket match 7, 21, 45, 12, 56, 35, 25, 0, 58, 66, 29. Find the median score.
18. The Median of the following data is 24. Find the value of x .

Class Interval (CI)	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50
Frequency (f)	6	24	x	16	9

19. Calculate the median from the following data:

Value	Frequency
0-10	4
10-20	12
20-30	24
30-40	36
40-50	20
50-60	16
60-70	8
70-80	5

20. The following figures relate to the preferences regarding screen size in inches of TV sets of 30 persons selected randomly from a locality. Find the modal size of the TV screen.

12 20 12 24 27 20 12 20 27 24 24 20 12 20
 24 27 24 24 20 24 24 20 24 24 12 24 20
 27 24 24

21. The following data relate to the sales of 100 companies:

Sales (Tk. lakhs)	No. of companies
Below 60	12
60-62	18
62-64	25
64-66	30
66-68	10
68-70	3
70-72	2

Calculate the modal sales.

22. The daily profits in TK's of 100 shops are given as follows:

Profits (in Tk's)	No. of shop
0-1000	12
1000-2000	18
2000-3000	27
3000-4000	20
4000-5000	17
5000-6000	6

Draw the histogram and then find the modal value. Check this value by direct calculation.

23. Find the mean, mode, and median for the following data:

Class	20-30	30-40	40-50	50-60	60-70
Frequency	8	16	40	30	6

Measure of Dispersion

24. Find the **standard deviation** from the weekly wages of ten workers working in a factory:

Workers	wages	workers	wages
A	320	F	340
B	310	G	325
C	315	H	321
D	322	I	320
E	326	J	331

25. An analysis of production rejects resulted in the following figures:

No. of rejects per operator	No. of operators
21-25	5
26-30	15
31-35	28
36-40	42
41-45	15
46-50	12
51-55	3

26. Calculate the coefficient of variation from the following data:

Profits (Rs. crores)	10-20	20-30	30-40	40-50	50-60
No. of Companies	8	12	20	6	4

27. A purchasing agent obtained samples of 60-watt bulbs from two companies. He had the samples tested in his laboratory for length of life with the following result:

Length of life (in hours)	Company A	Company B
1700-1900	10	3
1900-2100	16	40
2100-2300	20	12
2300-2500	8	3
2500-2700	6	2

- Which company's bulbs do you think are better?
- If prices of both types are the same, which company's bulbs would you buy?

28. There are 100 companies in an area. Within 100 there are a few numbers of companies selected randomly and their profit of thousand takas in a certain day is as follows: 50, 80, 70, 120, 200, 150, 130, 60. Find the CV.

Probability

29. A bag contains 4 white and 6 black balls. If one ball is drawn at random from the bag, what is the probability that the ball is

- Black - White - White or black	- Red - No red
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30. Tickets are numbered from 1 to 100. They are well shuffled and a ticket is drawn at random. What is the probability that the drawn ticket has

- An odd number
- A number 4 or multiple of 4
- A number which is greater than 70
- a square number?

31. Two dice are thrown together. What is the probability that the number obtained on one of the dies is a multiple of the number received on the other dice?

32. A box contains 5 red, 3 green, and 2 blue balls. Two balls are drawn at random. What is the probability that they are of different colors?

33. A box contains seven blue balls and five red balls, two balls are drawn at random without replacement, find the following:

- a. The probability that both balls are blue
- b. The probability that both balls are red
- c. The probability that one of the balls is blue
- d. The probability that at least one of the balls is blue
- e. The probability that at most one of the balls is red
- f. The probability that more than one ball is blue.

34. In a class of 20 students, 12 are girls. What is the probability of randomly selecting a boy?

35. In a class of 60 students, 30 are boys. If two students are selected randomly, what is the probability that both are girls?

36. From a standard deck of 52 cards, what is the probability of drawing a heart or a queen? From a deck of 52 cards, what is the probability of drawing an ace or a king?

37. Mr. X feels that the probability that he will pass Mathematics is $\frac{2}{3}$ and Statistics is $\frac{5}{6}$. If the probability that he will pass both the course is $\frac{3}{5}$. What is the probability that he will pass at least one of the courses?

38. Mr. Y feels that the probability that he will get A in Calculus is $\frac{3}{4}$, A in Statistics is $\frac{4}{5}$ and A in both the courses is $\frac{3}{5}$. What is the probability that Mr. Y will get the followings:

- a. at least one A
- b. no A 's?

39. Jessica has drawn a card from a well-shuffled deck and two cards at random from 52 cards. What is the probability of getting the following:

a. Two aces	f. Face card
b. Different colors	g. Heart of Kings
c. One black	h. Heart of Ace
d. Two Red	i. Spades
e. Jack of Diamond	